You are developing a **Product Management Component** in a MERN application.
Instead of fetching from backend, use a **local product array**.

Each product contains:

```
{
  id,
  name,
  price,
  category,
  stock
}
```

Requirements

useState()

Use useState() to:

- Store the product list
- Store search text
- Store selected category
- Store cart count
- Store cart items

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useRef()

Use useRef() to:

- Auto-focus the search input when component loads
- Store previous search value (without re-rendering)
- Scroll to product list when search button is clicked

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useMemo()

Use useMemo() to:

- Filter products based on:
  - Search text
  - Selected category
- Calculate total inventory value:

  total = sum(price × stock)

Ensure calculation only runs when dependencies change.

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useCallback()

- Create an addToCart function.
- Use useCallback() to memoize it.
- Pass it to a child component <ProductCard />.
- Wrap ProductCard using React.memo() to prevent unnecessary re-renders.

Given Local Data

```
const initialProducts = [
  { id: 1, name: "Laptop", price: 50000, category: "Electronics", stock: 5 },
  { id: 2, name: "Mobile", price: 20000, category: "Electronics", stock: 10 },
  { id: 3, name: "Chair", price: 3000, category: "Furniture", stock: 15 },
  { id: 4, name: "Table", price: 7000, category: "Furniture", stock: 8 }
];
```

Tasks

1. Display all products.
2. Add a search input field.
3. Add a category dropdown.
4. Show filtered products.
5. Display total inventory value.
6. Add "Add to Cart" button.
7. Show cart count.
8. Optimize performance using:
    - useMemo
    - useCallback
    - useRef
    - React.memo